

Lab Thirteen – Statistical Inference

1 Remaining Portfolio Work!

By 5p 4/27 Email me a link to your GitHub page with all polished labs rendered, and your review material (at least one unit). In the email, you should include a reflection about the processes involved in creating this portfolio and what you learned as a result. If you're having trouble reflecting, consider Fink's Taxonomy of Significant Learning (Google Fink taxonomy aligned self-reflection questions). In the email, let me know whether you would be comfortable sharing your page – I'm hoping that together we can get a good study guide for the final.

2 Statistical Inference

Kasdin et al. (2025) show that dopamine in the brains of young zebra finches acts as a learning signal, increasing when they sing closer to their adult song and decreasing when they sing further away, effectively guiding their vocal development through trial-and-error. This suggests that complex natural behaviors, like learning to sing, are shaped by dopamine-driven reinforcement learning, similar to how artificial intelligence learns. You can find the paper at this link: <https://www.nature.com/articles/s41586-025-08729-1>.

Note that the measurement is rather complicated. First, the measurements are taken on syllables of a bird song. When discussing bird songs, the term syllable refers to a single, distinct, uninterrupted sound that is part of a possibly complex bird song.

They initially describe their measurement as follows. They compare the sound of the developing zebra finches to the median sound of adult zebra finches, that act as a “goal” sound.

”For each syllable rendition, we calculated the proximity (Euclidean distance) to the median of the adult renditions of the syllable (age > 90 days) in feature space (Fig. 2a,b).”

They contextualize that measurement by computing a “relative quality” that takes into account their current ability (e.g., are they getting closer or further compared to their previous 14 renditions).

”Because dopamine is hypothesized to encode the relative value of performance, we defined the ‘relative quality’ of a syllable as proximity to the adult version of the syllable relative to the mean of the previous 14 renditions (Methods).”

They define the “closer” measurements as the 10% of syllable renditions with the highest relative quality and the “further” as the lowest 10%.

”To test for online error responses, for each day we compared the dopamine activity associated with the
10

Note that the data are measures of dopamine change, they use fibre photometry, changes in the fluorescence indicate dopamine changes in realtime. Their specific measurement considers the percent change in fluorescence in 100-ms windows between 200 and 300 ms from the start of singing, averaged across development.

Finally, the researchers note that they adjust the observations to correct for the increased dopamine for singing at all (whether it is close or far).

”As previously reported, the baseline level of dopamine in Area X is known to increase during singing; to isolate the effect of syllable quality from these singing-related dopamine increases, we subtracted the mean dopamine response across all renditions of a syllable on each day from the dopamine response for each rendition.”

So, the random variables we are working with are:

X_{close} = average % change in fluorescence for the top 10% syllable renditions
relative to distance from adult sound (corrected for singing)

X_{far} = average % change in fluorescence for the bottom 10% syllable renditions
relative to distance from adult sound (corrected for singing)

That is, the unit of measure is the syllable as measured on six birds, and the measurement is somewhat complicated.

2.1 Part a

Using the `pwr` package for R (Champely 2020), conduct a power analysis. How many observations would the researchers need to detect a moderate-to-large effect ($d = 0.65$) when using $\alpha = 0.05$ and default power (0.80) for a two-sided one sample t test.

Solution

Here, I can use the `pwr` package to conduct a `t.test` with the given inputs (“ d , α , and power”). The researchers would need at least 21 (can’t observe 20.58 birds) observations to see an effect size of $d = 0.65$.

```
1 power.t.test(d=0.65, sig.level = 0.05, power = 0.8, type = "one.sample", alternative = "two.sided")
```

One-sample t test power calculation

```
      n = 20.58044
delta = 0.65
      sd = 1
sig.level = 0.05
      power = 0.8
alternative = two.sided
```

2.2 Part b

Click the link to go to the paper. Find the source data for Figure 2. Download the Excel file. Describe what you needed to do to collect the data for Figure 2(g). Note that you only need the `closer_vals` and `further_vals`. Ensure to `mutate()` the data to get a difference (e.g., `closer_vals - further_vals`).

Solution

I downloaded the excel file, then grabbed the data from the `further_vals` and `closer_vals` tabs to manually copy paste into a more friendly csv file. Yes, I accidentally misspelled `further` as `farther`, and that shows an easy slip up in this process. After putting the data into a csv, I added it to my working folder, and read it using `tidyverse`. I also cleaned the data and added a new column that gets the difference between the `closer` and `farther` vals.

```
1 data = read_csv("fig_2g.csv")
2 data = data |>
3   select(closer_vals, farther_vals) |>
4   mutate(diff = closer_vals - farther_vals)
5 view(data)
```

2.3 Part c

Summarize the data.

- Summarize the further data. Do the data suggest that dopamine in the brains of young zebra finches decreases when they sing further away?

Solution

```

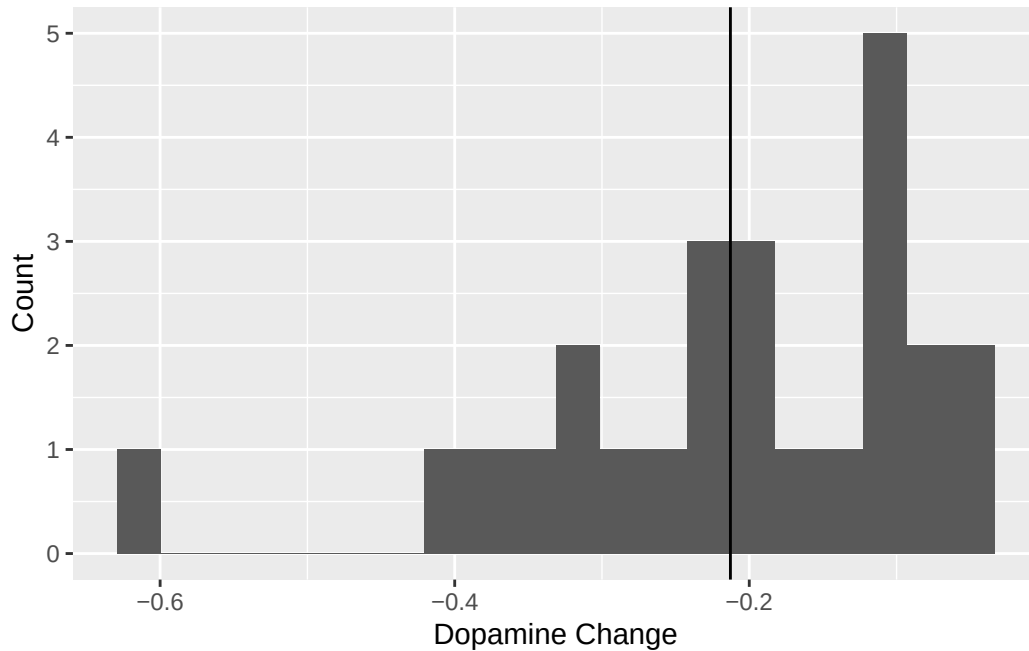
1 summary(data$farther_vals)

      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-0.6145 -0.2983 -0.1880 -0.2128 -0.1070 -0.0484

1 ggplot(data, aes(x = farther_vals)) +
2   geom_histogram(bins = 20) +
3   labs(x = "Dopamine Change",
4         y = "Count") +
5   geom_vline(xintercept = mean(data$farther_vals))

```

Figure 1: Dopamine for Farther Syllables



```

1 mean(data$farther_vals)

```

```
[1] -0.2128063
```

Here the data suggest that the dopamine decreases when the finches sing further away. This is seen in the average negative dopamine response.

- Summarize the closer data. Do the data suggest that dopamine in the brains of young zebra finches increases when they sing closer to their adult song?

Solution

```

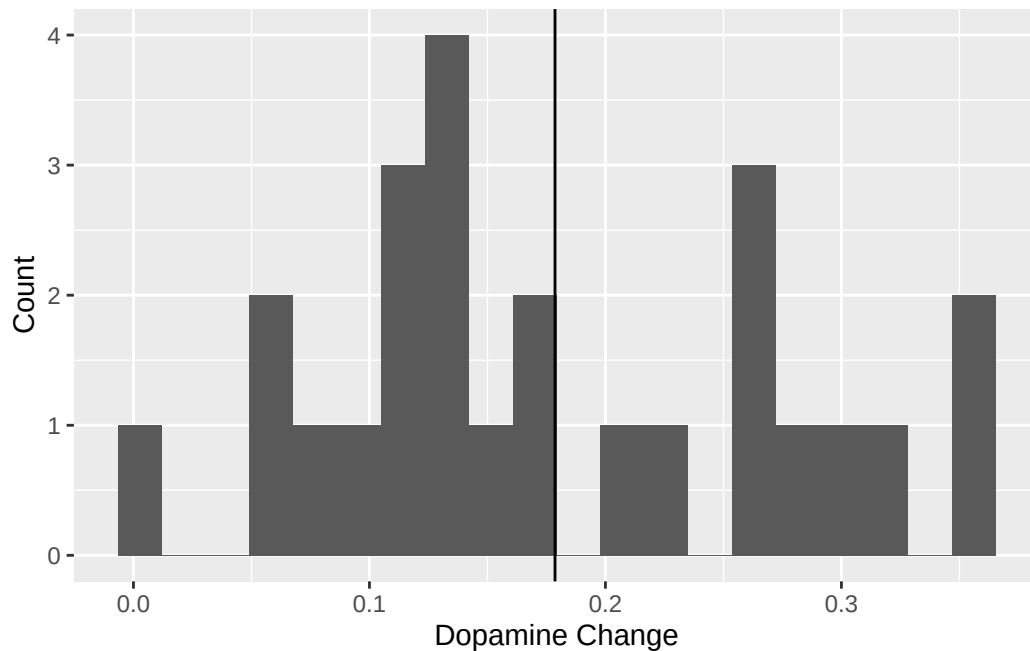
1 summary(data$closer_vals)

      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.002665 0.120542 0.143664 0.178624 0.263011 0.355983

1 ggplot(data, aes(x = closer_vals)) +
2   geom_histogram(bins = 20) +
3   labs(x = "Dopamine Change",
4         y = "Count") +
5   geom_vline(xintercept = mean(data$closer_vals))

```

Figure 2: Dopamine for Closer Syllables



```
1 mean(data$closer_vals)
```

```
[1] 0.1786241
```

Here the data suggest that dopamine increases on average when the finches sing closer. The mean is positive here.

- Summarize the paired differences. Do the data suggest that there is a difference between dopamine in the brains of young zebra finches when they sing further away compared to closer to their adult song?

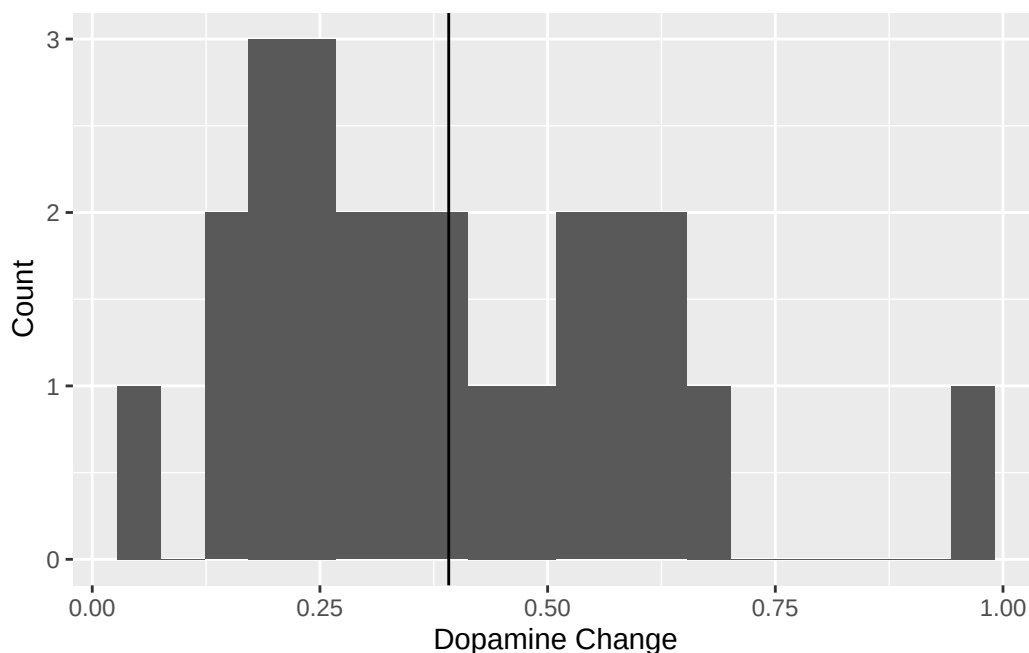
Solution

```
1 summary(data$diff)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.05106 0.22831 0.35330 0.39143 0.53337 0.96657
```

```
1 ggplot(data, aes(x = diff)) +
2   geom_histogram(bins = 20) +
3   labs(x = "Dopamine Change",
4        y = "Count") +
5   geom_vline(xintercept = mean(data$diff))
```

Figure 3: Difference (closer-farther) in Dopamine



```
1 mean(data$diff)
```

```
[1] 0.3914303
```

Here, because the mean is positive, the data suggest that the closer songs always gives greater dopamine compared to the farther songs.

- **Optional Challenge** Can you reproduce Figure 2(g)? Note that the you can use `geom_errorbar()` to plot the range created by adding the mean $\pm$ one standard deviation.

Solution

I'm okay thanks.

2.4 Part d

Conduct the inferences they do in the paper. Make sure to report the results a little more comprehensively – that is your parenthetical should look something like: ($t = 23.99$, $p < 0.0001$; $g = 1.34$; 95% CI: 4.43, 4.60). Ensure to verify any conditions for use.

Note: Your numbers will vary slightly because they reported two-sided test p -values instead of the specified one-sided tests.

- “The close responses differed significantly from 0 ($p = 1.63 \times 10^{-8}$).”

Note: Use the appropriate one-sided test.

Solution

I get to cheat and use `t.test` so all of this is mostly trivial.

```
1 t_close = t.test(data$closer_vals, mu = 0, alternative = "greater")
2 t_close
```

One Sample t-test

```
data: data$closer_vals
t = 9.1573, df = 24, p-value = 1.333e-09
```

```
alternative hypothesis: true mean is greater than 0
95 percent confidence interval:
 0.1452511      Inf
sample estimates:
mean of x
0.1786241
```

The dopamine response for closer singing is significantly greater than 0 ($t = 9.16$, $p < 0.0001$, 95% CI: 0.145, ∞), suggesting dopamine increases when birds sing closer.

- “The far responses differed significantly from 0 ($p = 5.17 \times 10^{-8}$).”

Note: Use the appropriate one-sided test.

Solution

```
1 t_far = t.test(data$farther_vals, mu = 0, alternative = "less")
2 t_far
```

One Sample t-test

```
data: data$farther_vals
t = -7.9245, df = 24, p-value = 1.866e-08
alternative hypothesis: true mean is less than 0
95 percent confidence interval:
 -Inf -0.1668619
sample estimates:
mean of x
-0.2128063
```

The dopamine response for farther singing is significantly less than 0 ($t = -7.92$, $p < 0.0001$, 95% CI: $-\infty$, -0.167), suggesting dopamine decreases when birds sing farther.

- “The difference between populations was significant ($p = 1.04 \times 10^{-8}$).”

Note: Use the two-sided test.

Solution

```
1 t_diff = t.test(data$diff, mu = 0, alternative = "two.sided")
2 t_diff
```

One Sample t-test

```
data: data$diff
t = 9.1428, df = 24, p-value = 2.746e-09
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 0.3030683 0.4797923
sample estimates:
mean of x
0.3914303
```

The paired difference between closer and farther singing is statistically significant ($t = 9.14$, $p < 0.0001$, 95% CI: 0.303, 0.480), signifying a statistical difference in dopamine responses between singing close and far.

2.5 Part e

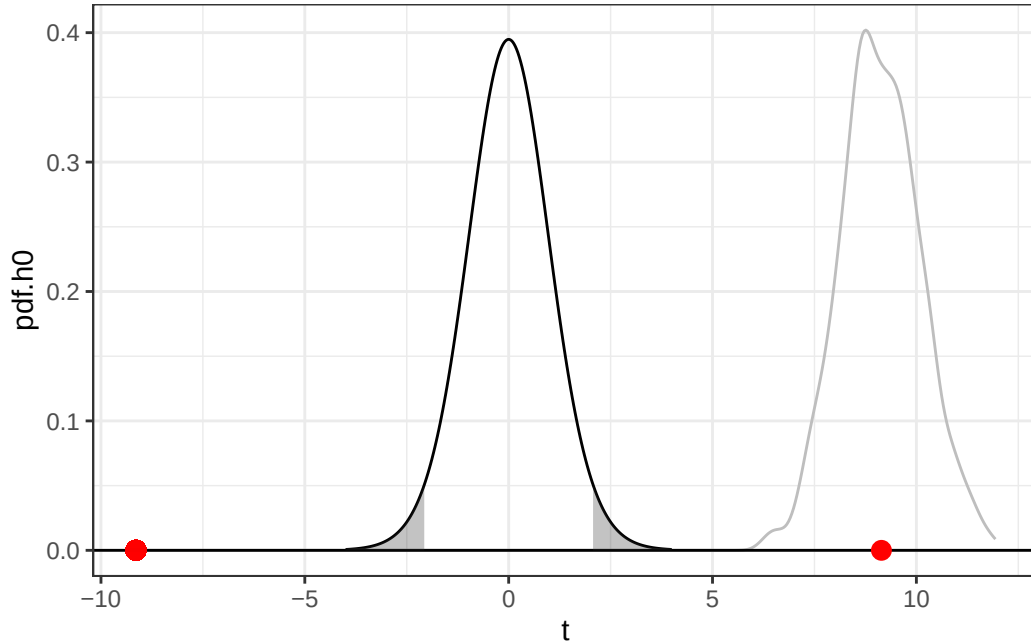
For each hypothesis test in the previous question, plot the null T distribution and the observed t statistic. Ensure to use `geom_ribbon()` to highlight the rejection region and the p -value. Additionally, add the resampling distribution for T to the chart to demonstrate any differences between the null distribution and the distribution suggested by the data.

```

1  set.seed(7272)
2
3  t.stat = t_diff$statistic
4  n = length(data$diff)
5  s = sd(data$diff)
6  mu0 = 0
7
8  #null t distr
9  ggdat = tibble(t = seq(-4, 4, length.out = 1000)) |>
10    mutate(pdf.h0 = dt(t, df = n-1))
11
12  #resampling
13  ts = rep(NA, 1000)
14  for(i in 1:1000){
15    curr_sample = sample(data$diff, size=n, replace=T)
16    ts[i] = (mean(curr_sample) - mu0)/(s/sqrt(n))
17  }
18
19  #rejection zone
20  alpha = 0.05
21  t_crit = qt(1 - alpha/2, n-1)
22
23  ggplot(ggdat) +
24    geom_line(aes(x=t, y=pdf.h0))+
25    stat_density(data=tibble(t=ts), aes(x=t), color="grey", geom="line")+
26    geom_hline(yintercept=0) +
27    # Plot observation
28    geom_point(data=tibble(x=t.stat, y=0),
29              aes(x=x, y=y), color="red", size=3) +
30    # Plot mirrored observation
31    geom_point(aes(x = -t.stat, y = 0),
32              color = "red", size = 3, alpha = 0.3) +
33    # Rejection zone
34    #Right tail
35    geom_ribbon(data = ggdat
36              |> filter(t>= t_crit),
37              aes(x = t, ymin = 0, ymax = pdf.h0),
38              alpha = 0.3) +
39    #Left tail
40    geom_ribbon(data = ggdat
41              |> filter(t<= -t_crit),
42              aes(x = t, ymin = 0, ymax = pdf.h0),
43              alpha = 0.3) +
44
45    theme_bw()+
46    xlab("t")+
47    ylab("pdf.h0")

```

Figure 4: T distribution with rejection region shown



Because our observed t statistic is well into the rejection region of the null distribution, this signifies the observation is very unlikely ($p=2.74 \text{ e-}09$) under the null hypothesis. Also, the resampling distribution is centered away from 0, thus we can say the observed effect is not due to random chance either. Therefore, the alternative hypothesis has statistical support that there is a true difference in dopamine response in the paired difference of singing close versus far from the real song.

Genuinely thank you for a great semester. Statistics is by far my weakest mathematical area, so I came into this class with very low expectations. You pushed me a lot harder and I came to appreciate this topic for which I had ignorantly passed judgment. Thank you for being funny (yes ur dumb stats jokes somehow make me chuckle) and I hope you have a great summer.

P.S. Please mess with Oscar as much as you can next year. Poor guy, who's gonna give him a hard time if I'm not there anymore?

References

- Champely, Stephane. 2020. *Pwr: Basic Functions for Power Analysis*. <https://CRAN.R-project.org/package=pwr>.
- Kasdin, Jonathan, Alison Duffy, Nathan Nadler, et al. 2025. "Natural Behaviour Is Learned Through Dopamine-Mediated Reinforcement." *Nature*, 1–8.